Hands-on Introduction to Computer Architecture with RISC-V

DESIGN OF A PROGRAMMABLE STATE MACHINE

ASSIGNMENT 02

NAME : SHANIKA KAVINDI

REG.NO : SKF 2500222

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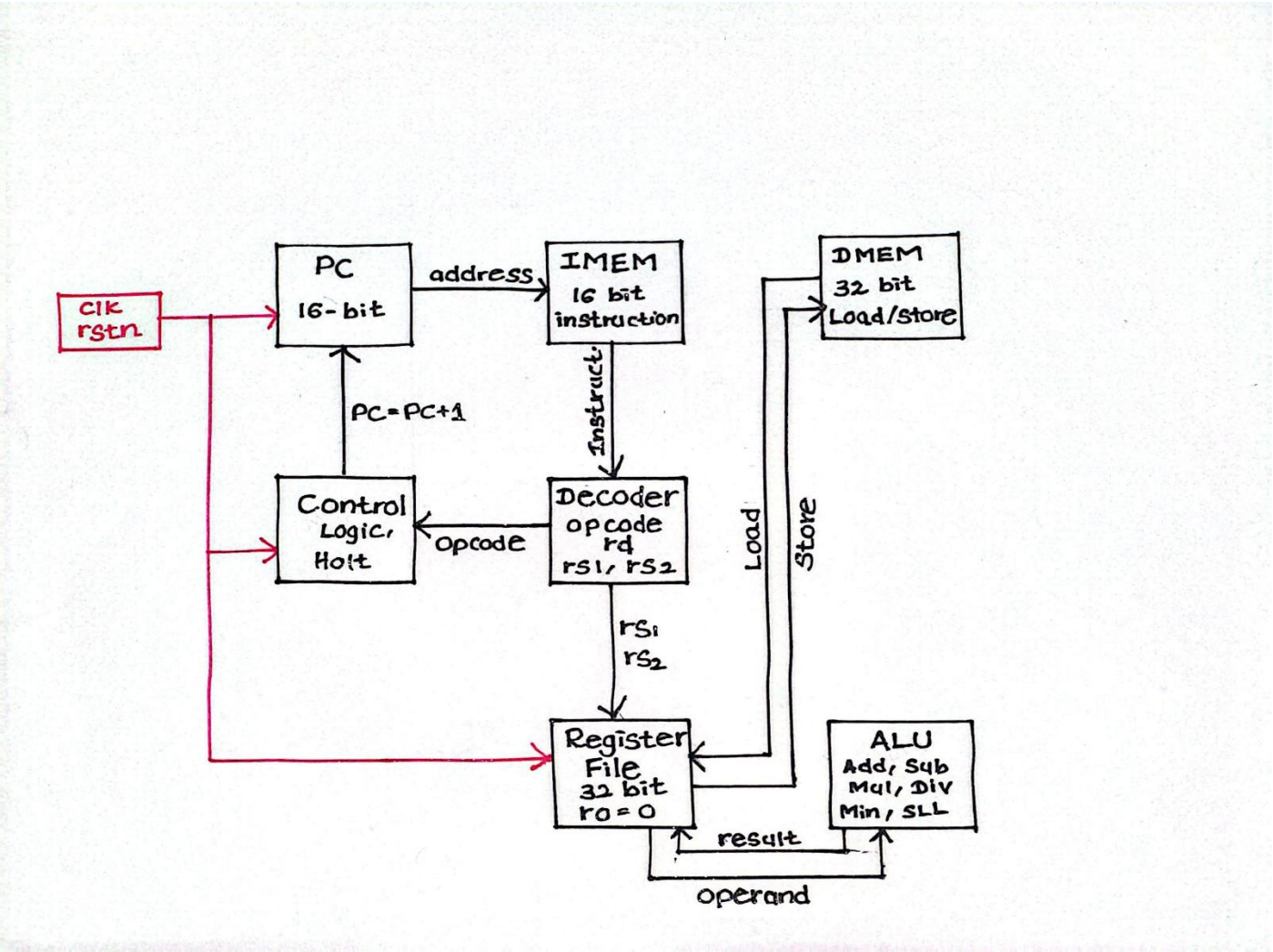
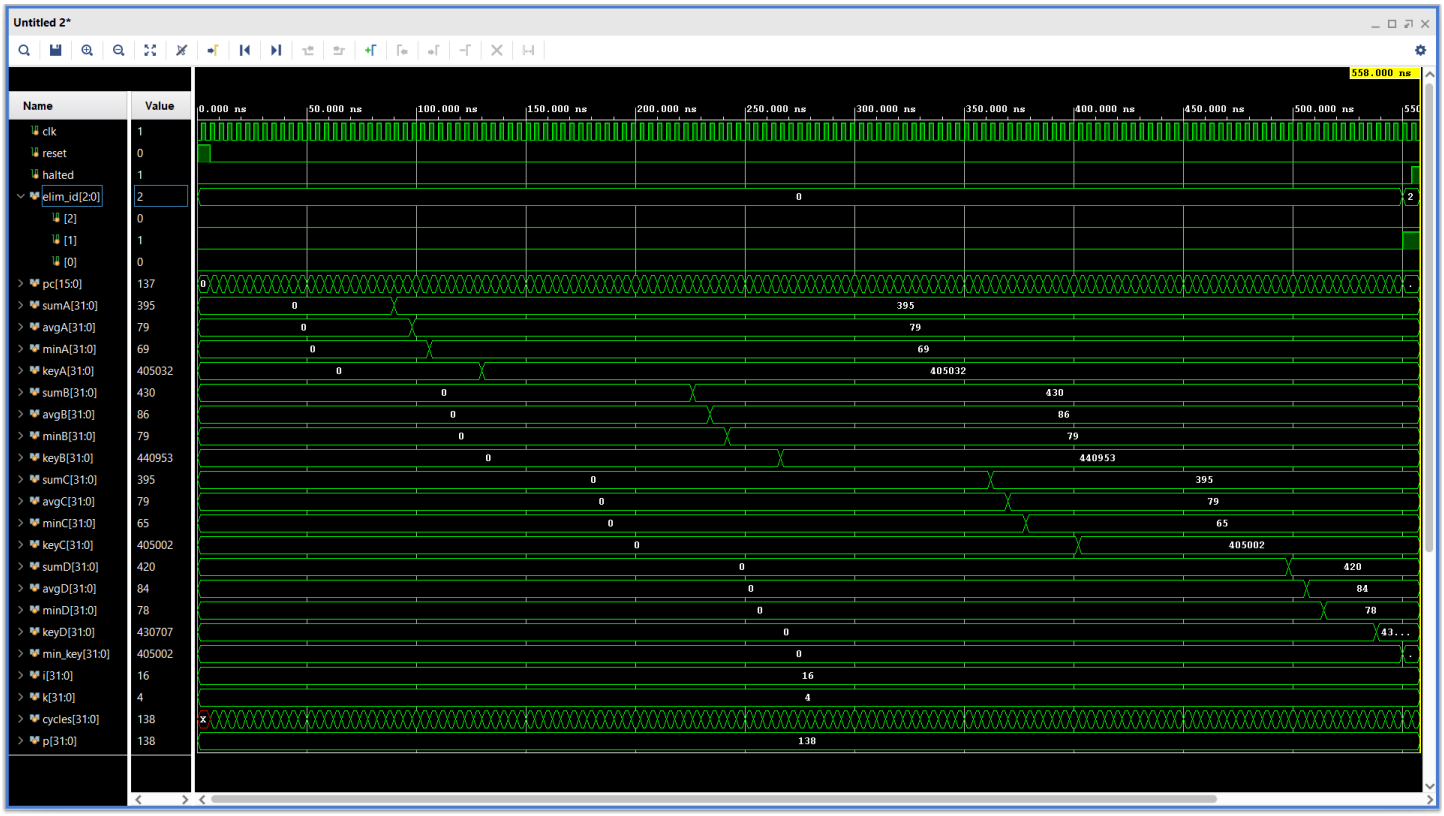
Introduction

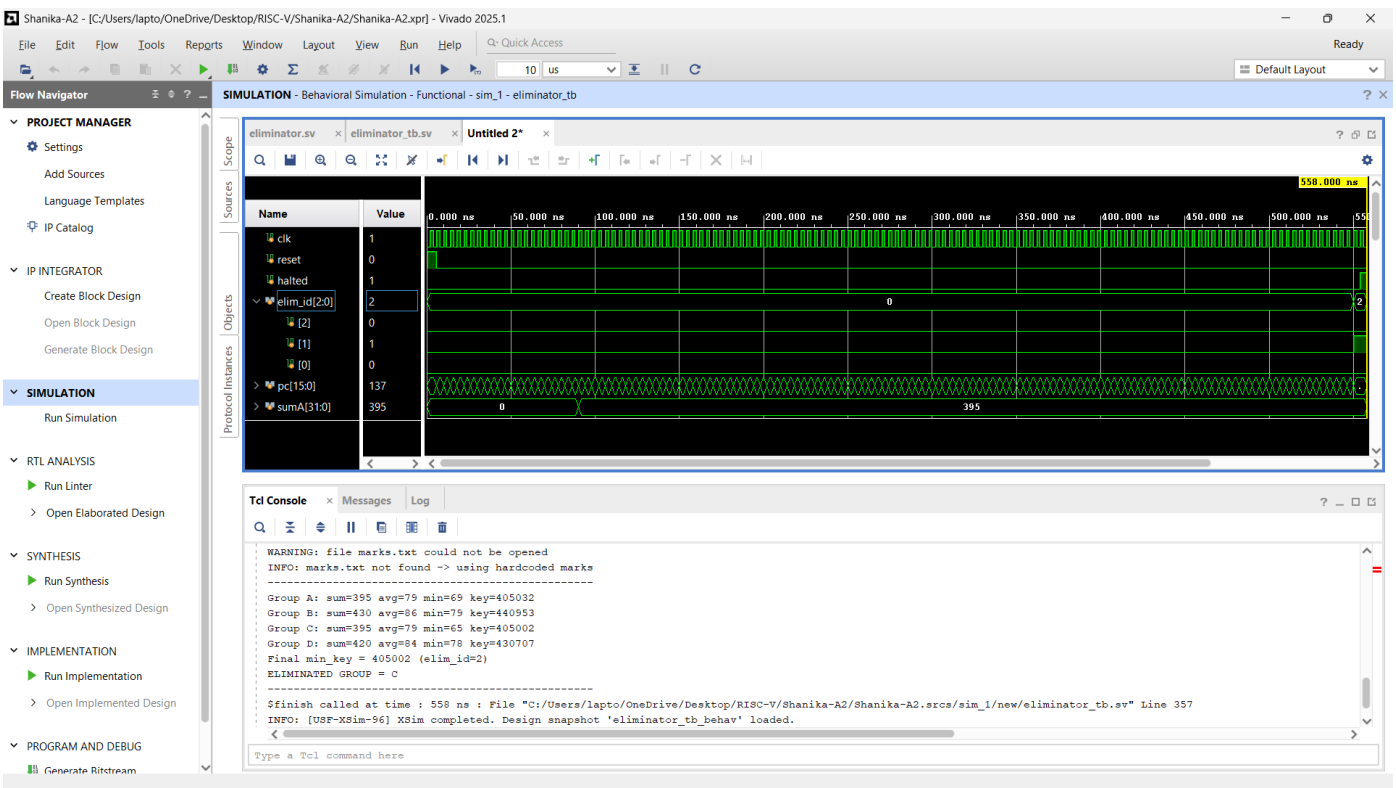
This report presents the design and implementation of a programmable state machine CPU to solve a student elimination problem. The system processes test scores from 20 students divided into 4 groups (A, B, C, D) of 5 students each. The CPU computes each group's average score, identifies the group with the lowest average, and applies a tie-breaking rule based on minimum individual scores. The design utilizes a custom 16-bit instruction set architecture with 16 registers, instruction memory (IMEM), and data memory (DMEM), executing one instruction per clock cycle.

System block diagram



Results and waveforms





Conclusion

The programmable state machine successfully executed the elimination algorithm and processed all 20 student marks. Based on the given test data, the system computed the following results:

- Group A: Average = 79, Minimum = 69
- Group B: Average = 86, Minimum = 79
- Group C: Average = 79, Minimum = 65
- Group D: Average = 84, Minimum = 78

Eliminated Group: C

Group C was eliminated due to having the lowest average score (79) tied with Group A, but containing the lowest individual mark (65) among all tied groups. The design demonstrates effective use of a custom instruction set including arithmetic operations (ADD, SUB, MUL, DIV), comparison (MIN), and memory operations (LOAD, STORE) to solve real-world computational problems. The simulation verified correct operation with the halted signal activating upon program completion and all intermediate calculations stored properly in data memory.